

CAN DEVELOPMENTAL PLASTICITY RESCUE ECTOTHERMS FROM RISING TEMPERATURES?

Mapping the current state of knowledge



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BACKGROUND

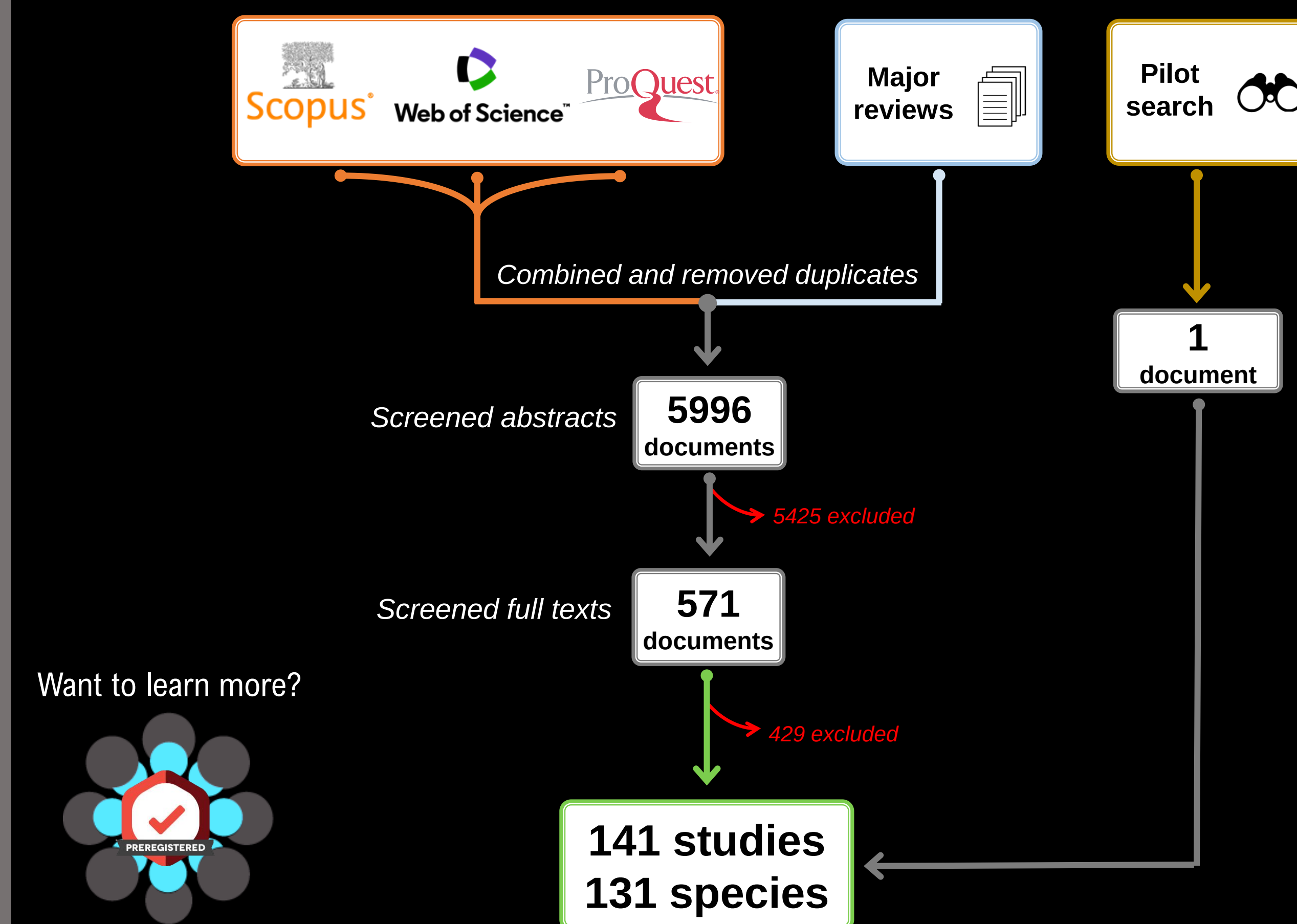
- Rising temperatures may expose ectotherms to temperatures beyond their physiological limits
- These animals can adjust their capacity to withstand heat by acclimating to new temperatures
- Temperatures experienced during development can profoundly influence later phenotypes
- However, most published reviews focused on the acclimation of adults, leaving little emphasis to early developmental plasticity

AIMS

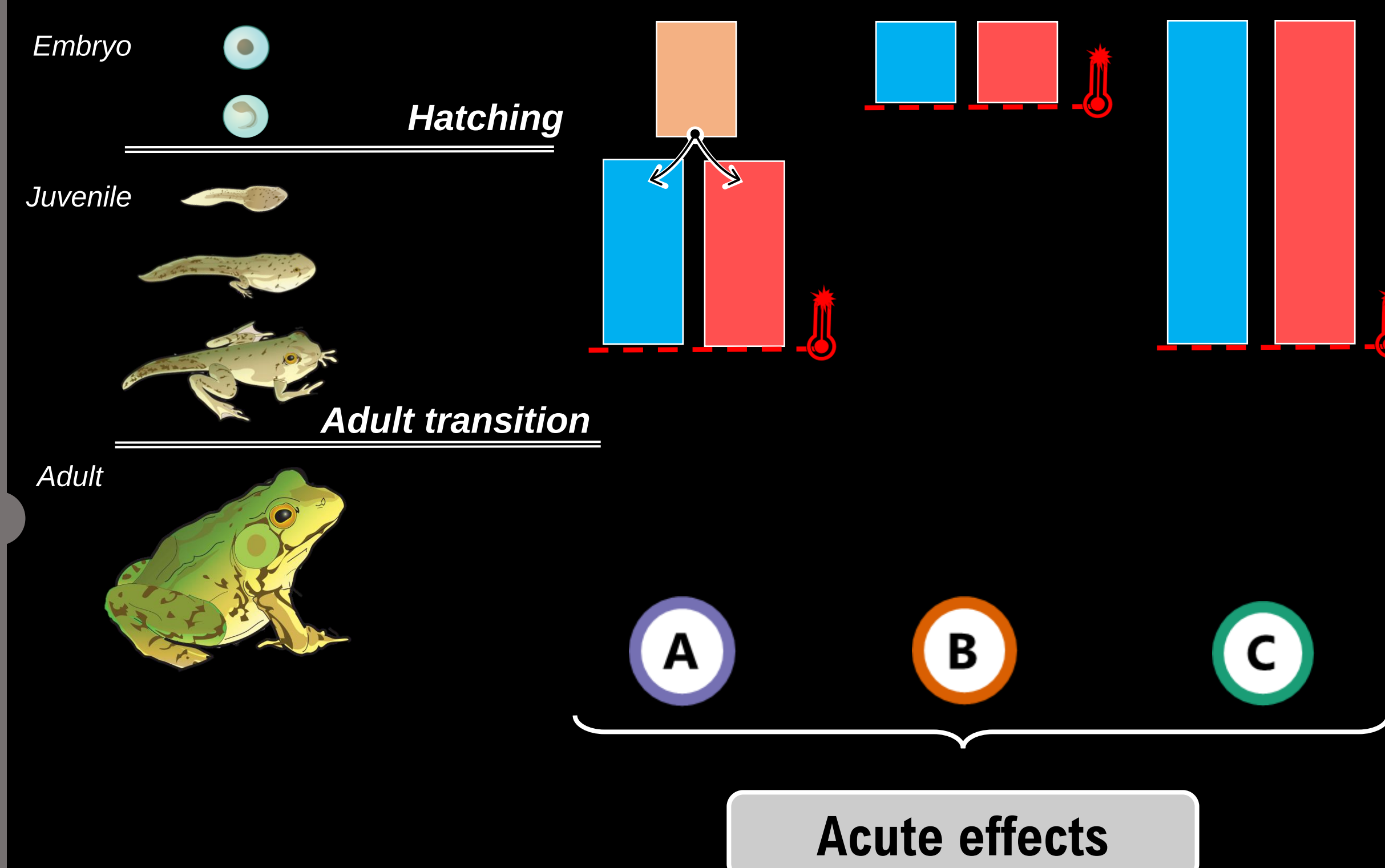
- Synthesise studies testing the developmental plasticity of heat tolerance
- Map the evidence to identify knowledge-gaps and methodological considerations
- Compile and meta-analyse the published data (*in progress*)

METHODS

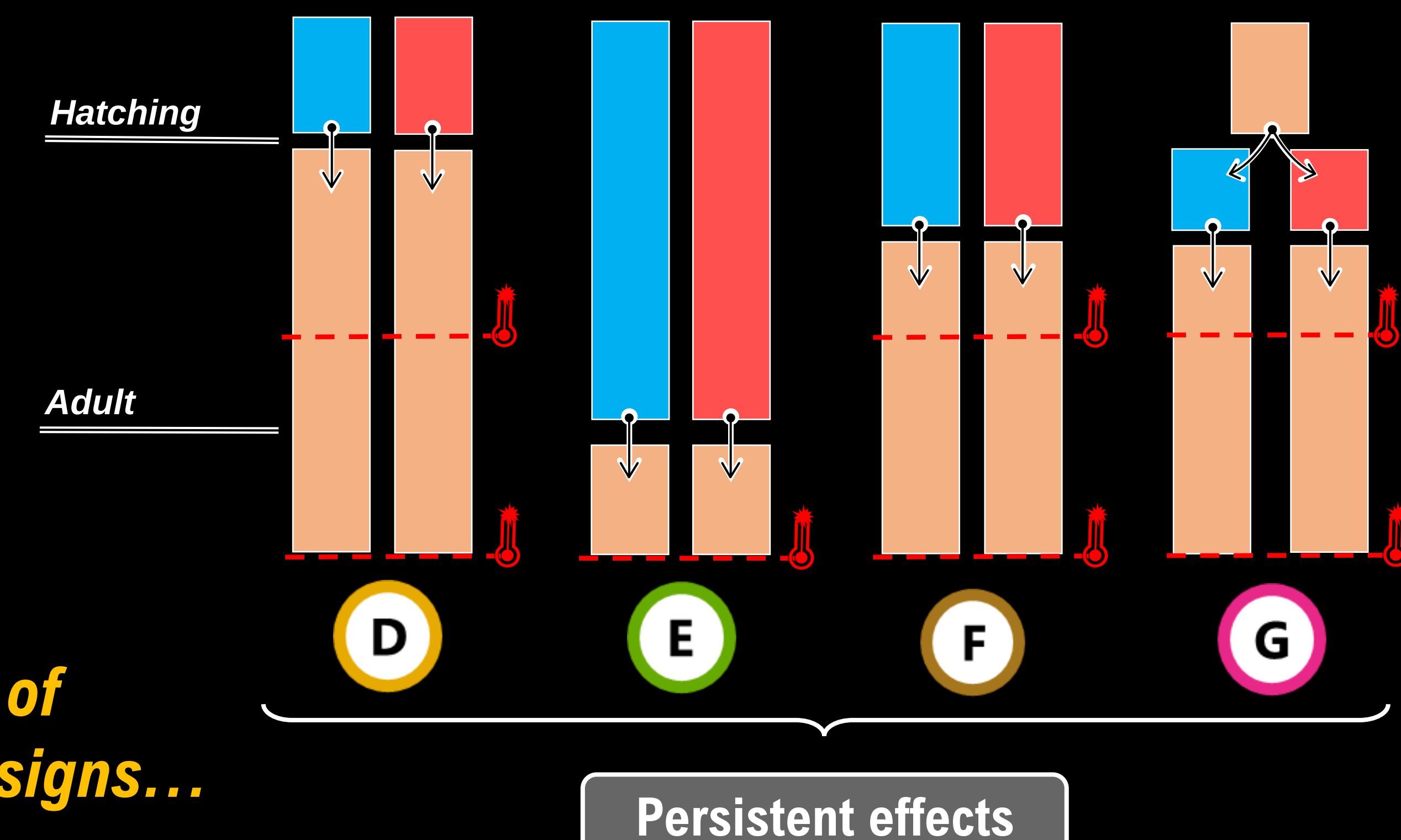
- Systematic review
 - ✓ Ectotherm
 - ✓ Heat tolerance
 - ✓ ≥ 2 developmental temperatures



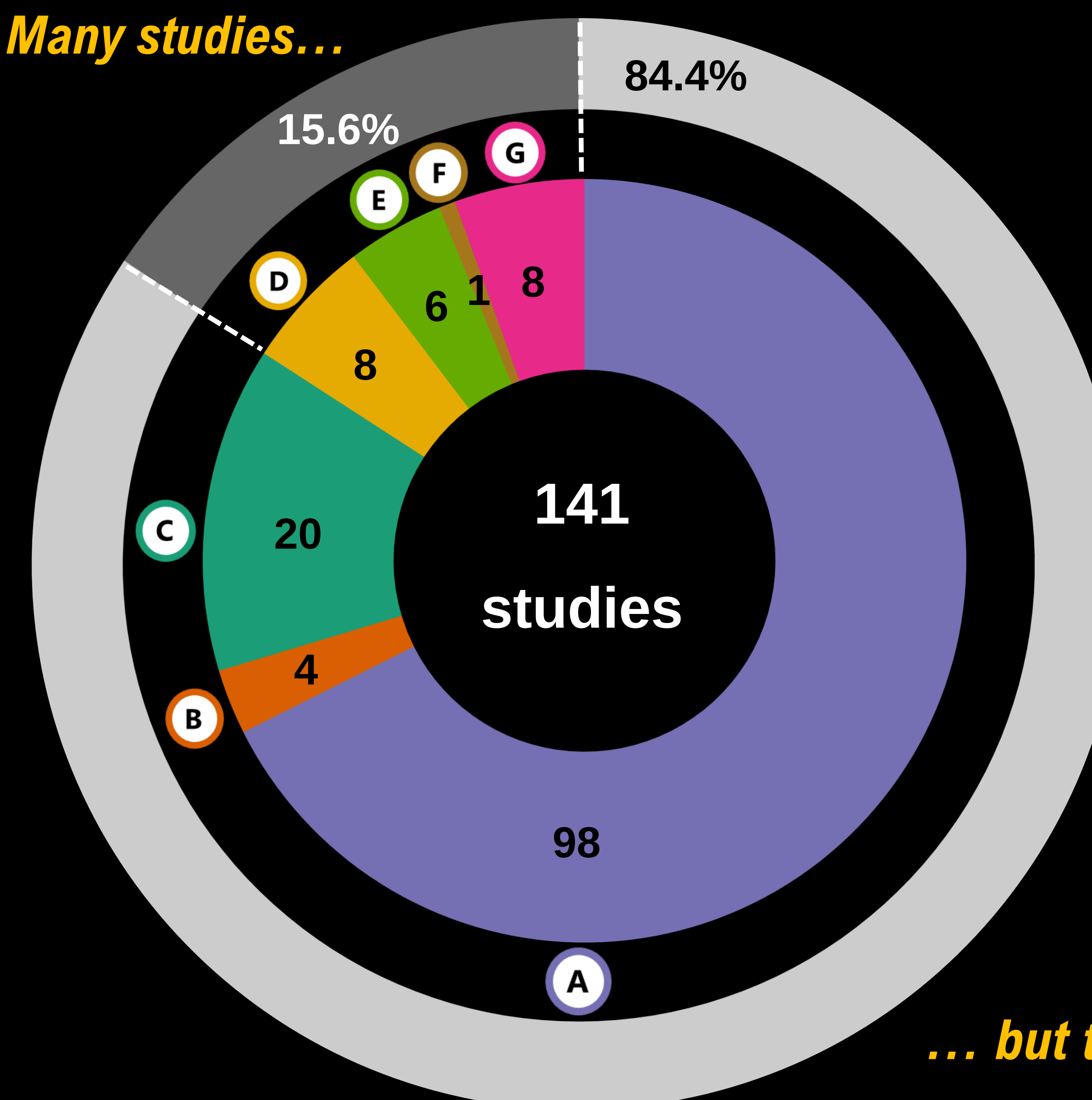
THE CURRENT STATE OF KNOWLEDGE



A diversity of experimental designs...

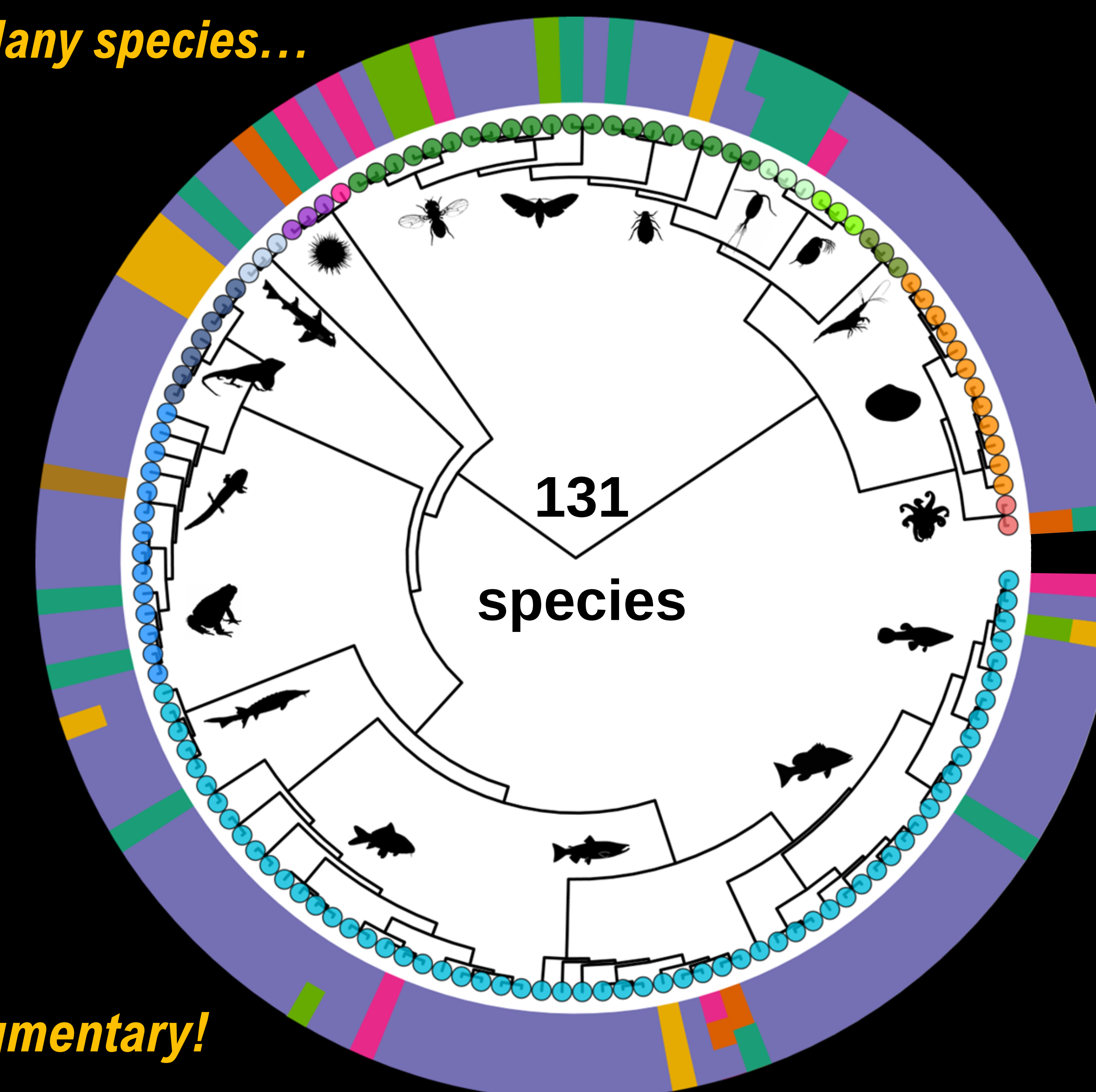


Many studies...



Many species...

● Insecta
● Hexanauplia
● Branchipoda
● Malacostraca
● Bivalvia
● Cephalopoda
● Actinopterygii
● Amphibia
● Reptilia
● Chondrichthyes
● Echinoidea
● Holothuroidea



... but the evidence is fragmentary!

ACKNOWLEDGEMENTS



KEY FINDINGS

- Research dominated by fish species
- Huge diversity of experimental designs
- The persistence of developmental effects is rarely tested
- Knowledge-gaps remain in most taxa

WHAT IS NEXT?

- More research on the persistence of developmental effects
- More research on the heat tolerance of embryos
- Standardize methods in experimental research
- Quantitatively analyse the data (*in progress*)